HINTS & SOLUTIONS OF WAVE ON A STRING

EXERCISE-1
PART – I

A-1. (a) amplitude — A = 10 mm

(b) wave number — $k = 5\pi$ cm^{-1}

(c) wavelength — $\lambda = \frac{2\pi}{k} = \frac{2}{5}$ cm

(d) frequency — $f = \frac{\omega}{2\pi} = 30$ Hz

(e) time period — $T = \frac{1}{f} = \frac{1}{30}$ s

(f) wave velocity — $u = f\lambda = 12$ cm/s.

A-2. $A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos \Delta\phi$

$\Delta\phi = \frac{\pi}{2}$, $A_1 = Y_m$, $A_2 = Y_m$

$Y^2 = Y_m^2 + Y_m^2 + 2Y_m^2 \times \cos \frac{\pi}{2}$

$Y^2 = Y_m^2 + Y_m^2$ $[Y = \sqrt{2}\, Y_m = 1.41\, Y_m]$

A-3. (a) $\omega = 2\pi f = 10\pi$ Red/sec

(b) $\lambda \times 5 = 20$ $\lambda = 4$ m $K = \frac{2\pi}{\lambda} = \frac{\pi}{2}$ rad/m

(c) $y = 12 \times 10^{-2} \sin(\omega t - kx + \phi)$

$= 12 \times 10^{-2} \sin(10\pi t - \frac{\pi}{2} x + \phi)$

At t = 0 x = 0 & y = 0

$0 = \sin\phi$

$\therefore$ $\phi = 0$ or π

$\frac{\partial y}{\partial x} = 12 \times 10^{-2} \left(-\frac{\pi}{2}\right) \cos(100\pi t - \frac{\pi}{2} x + \phi)$

At t = 0 पर x = 0

$\frac{\partial y}{\partial x} = -12 \times 10^{-2} \left(\frac{\pi}{2}\right) \cos\phi$

$\frac{\partial y}{\partial x}$ should be positive

$\therefore$ $\phi = \pi$

$y = 12 \times 10^{-2} \sin(10\pi t - \frac{\pi}{2} x + \pi)$

$= 12 \times 10^{-2} \sin(\frac{\pi}{2} x - 10\pi t)$

(d) $V_{max} = A\omega$

$= 12 \times 10^{-2} \times 10\pi$

$= 1.2\pi = \frac{6}{5}\pi$

(e) $A_{max} = A\omega^2 = 12 \times 10^{-2} \times 10\pi \times 10\pi$

$= 12\pi^2$

A-4. $\omega = \frac{2\pi}{T} = \frac{2\pi}{8-2} = \frac{\pi}{3}$

$\therefore \quad K = \frac{\omega}{v} = \frac{\pi}{3\times 6} = \frac{\pi}{18}$

$\therefore \quad y = 0.5 \sin\left[\frac{\pi}{3}t - \frac{\pi}{18}x + \phi\right]$

at t = 2, x = 8 , y = 0

$\therefore \quad \phi = \frac{7\pi}{9}$

$\therefore \quad y = 0.5 \sin\left[\frac{\pi}{3}t \quad -\frac{\pi}{18} \quad x + \frac{7\pi}{9}\right]$

A-5. (a) D, E, F $\Rightarrow$ (slope = –ve $\Rightarrow$ v_p = +ve)
(b) A, B, H $\Rightarrow$ (slope = +ve $\Rightarrow$ v_p = –ve)
(c) At C, G, [slope = 0]
(d) At A, E [slope is maximum]

B-1. $V = \sqrt{\frac{T}{\mu}} = \left(\frac{1350}{5\times 10^{-3}}\right)^{1/2} = 300\ \sqrt{3}\ \frac{m}{sec}$

B-2. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20\times 2.5}{5\times 10^{-3}}} = 100$ m/sec

$t = \frac{D}{v} = \frac{2}{100} = \frac{1}{50}$ sec

B-3. At the bottom end

$V = \sqrt{\frac{mg}{\mu}} = f\lambda$ .....(1)

Now at the top

$V_1 = \sqrt{\frac{mg + \mu\ell g}{\mu}} = f\lambda'$ .....(2)

$\mu\ell$ = M = mass of string
From equation (1) & (2)

$\lambda' = \sqrt{\frac{m+M}{m}}\lambda = \sqrt{\frac{2+8}{2}}\ \lambda = \sqrt{5}\ \lambda = \frac{3}{10\sqrt{5}}$ m

B-4. T/2 = 5 m sec. (T is time period)
T = 10 m sec.

$\therefore f = \frac{1}{10}$ (m sec.)$^{-1}$ = 100 sec^{-1}

$\frac{\lambda}{2}$ = 2 cm, λ = 4 cm ,

$v = \lambda f = 100 \times 0.04 = 4$ m/sec

B-5. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{A\rho}}$

$\frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \frac{1}{2}$

$\frac{\rho_2}{\rho_1} = \frac{1}{4}$

$\frac{\rho_1}{\rho_2} = 4$

B-6. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{4\times(10+6)}{1.6\times10^{-3}}} = 200$ m/sec

$V_{pully,\ ground} = V_{p,string} + V_{string,\ ground} = 200 + 40 = 240$ m/s

C-1. (a) f = 50Hz

$\mu = \frac{180\times10^{-3}}{6} = 3 \times 10^{-2}$ Kg/m

$2A = 15 \times 10^{-2}$ m

$A = 7.5 \times 10^{-2}$ m

$4\lambda = 6 \quad \lambda = 3/2$ m and $v = f\lambda = 75$ m/sec

$\omega = 2\pi f = 100\pi$ also $K = \frac{\omega}{v} = \frac{4\pi}{3}$

If phase constant is ϕ then

$\therefore$ Equation is $y = 7.5 \times 10^{-2} \sin\left[100\pi t - \frac{4\pi}{3}x + \phi\right]$

(b) $P_{av} = 2\pi^2 f^2 A^2 \mu v = \frac{2025}{32}\pi^2 \simeq 625$ w

C-2. $P_{av.} = 2\pi^2 f^2 A^2 \mu V$

$P_{av.} = 2\pi^2 f^2 A^2 \mu \sqrt{T/\mu}$

Put value $P_{av} = 50$ mw

use $v = \sqrt{T/\mu}$

D-1. Equation of reflected wave is

$y = \left(\frac{80a}{100}\right) \sin \frac{2\pi}{\lambda}\left(\upsilon t + x + \frac{\lambda}{2}\right)$

($\frac{\lambda}{2}$ path difference is added as the reflection is from hard surface)

(As the surface is Rigid)

D-2. (a) $a_{Net_{तुल्य}} = 0$

(b) $a_{Net_{तुल्य}} = a_1 + a_2 = 0.15 + 0.15 = 0.3$

D-3. $V = f\lambda$

$V = 100 \times 2 = 200 \frac{cm}{sec}$

(a) $\frac{2\pi}{T} \times \Delta t = \frac{2\pi\times0.01}{0.01} = 2\pi$ (b) $\phi = \frac{2\pi}{\lambda}\ \Delta x = \frac{2\pi}{2} = \pi$

(c) $a = a_1 + a_2 = 4$ mm

$a = a_1 - a_2 = 0$

E-1. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{25}{\frac{100\times10^{-3}}{10}}} = 50$

$\therefore \nu_0 = \frac{v}{2L} = \frac{50}{2\times10} = 2.5$ Hz

$\therefore \nu_1 = 2\nu_0 = 5$ Hz

$\nu_2 = 3\nu_0 = 7.5$ Hz

E-2. (a) $V = \sqrt{\frac{150}{7.2\times10^{-3}}} = \frac{250}{\sqrt{3}}\frac{m}{sec} \simeq 144$ m/s

(b) $\frac{3\lambda}{2} = 90$ cm $\Rightarrow \lambda = 60$ cm

(c) $V = f\lambda \Rightarrow f = \frac{v}{\lambda} = \frac{250\times100}{\sqrt{3}\times60} = \frac{1250}{3\sqrt{3}}$ Hz

E-3. $\frac{4\lambda}{2} = 1.5$

$\lambda = 0.75$ m

$v = \sqrt{\frac{10\times10\times1.5}{15\times10^{-3}}} = 100$ m/s

$f = \frac{v}{\lambda} = \frac{100}{0.75} = \frac{400}{3}$ Hz

E-4. (a) $A_{s\omega} = 0.5 \sin \frac{\pi}{3} x$

$= 0.5 \sin \frac{\pi}{3} (1.5) = 0.5$ cm

But the amplitude of component wave is A

(b) $2A = 0.5$

$A = 0.25$ cm

$K = \frac{\pi}{3}$ and $\omega = 40\pi \Rightarrow f = 20$ Hz

$\therefore V = \frac{\omega}{K} = 120$ cm/sec

(c) $d = \frac{\lambda}{2} = \frac{V}{2f} = 3$ cm

(d) At $x = 1.5$ cm

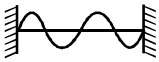
At $t = \frac{9}{8}$ sec

$y = 0.5 \sin\left(\frac{\pi}{3}\times1.5\right) \cos 40\pi \times \frac{9}{8}$

$= 0.5 \sin \frac{\pi}{2} \cos 45\,\pi$

$= 0.5 (-1) = -0.5$

So partical is at negative extreme position that is why speed is zero

E-5. (a) 

$2\lambda = \ell$

$4f_0 = 400 \quad \Rightarrow \quad f_0 = 100$ Hz

(b) 

$\frac{7\lambda}{2} = \lambda \quad \Rightarrow \quad \lambda = \frac{2\ell}{7}$

$\frac{v}{f} = \frac{2\ell}{7} \quad \therefore \quad f = 7\ \frac{v}{2\ell} = 700$ Hz

E-6. (i) $y_1 = 1.5 \cos\left(\frac{\pi x}{20} + 72\pi t\right)$

$y_2 = 1.5 \cos\left(\frac{\pi x}{20} - 72\pi t\right)$

(ii) Nodes निस्पंद $\Rightarrow$

$\frac{\pi x}{20} = (2n + 1)\ \frac{\pi}{2}$

$x = (2n + 1)\ 10 \quad \Rightarrow \quad x = 10, 30, 50.....$

(iii) Antinodes प्रस्पंद $\Rightarrow \quad \frac{\pi x}{20} = n\pi \Rightarrow x = 20\ n \Rightarrow x = 0, 20, 40, 60.....$

E-7. As end A is free so antinode is formed

$\therefore \quad \ell = \frac{\lambda_1}{4}$

Now if support is pushed to right by 10 cm so that node is formed at joint

$\therefore \quad \ell = \frac{\lambda}{2} \Rightarrow \left(\lambda_2 = \frac{\lambda_1}{2}\right)$

As V is same so if wavelength is halfed frequeny will be doubled i.e. 240 Hz

PART - II

A-1. As $\frac{5\lambda}{2} = 20 \quad \Rightarrow \quad \lambda = 8$ cm

$K = \frac{2\pi}{\lambda} = \frac{314}{4}$

$\omega = KV\ \frac{2\pi}{8\times10^{-2}} \times 350 = 27475$

$\therefore y = 0.05 \sin\left(\frac{314}{4}x - 27475t\right)$

A-2. $y = \frac{1}{1+x^2}$ (t = 0)

$y = \frac{1}{1+(x-2v)^2}$ (t = 2)

Now comparing

$x - 2v = x - 1$

$v = 0.5\ \frac{m}{sec}$

A-3. $V_{P_{max}} = A\omega = Y_0\, 2\pi f = 4V_\omega$

$Y_0\, 2\pi f = 4\left(\frac{2\pi f}{2\pi/\lambda}\right) \quad \therefore \quad \lambda = \frac{\pi Y_0}{2}$

A-4. Put α, β, A, x and t in the equation

$\frac{2\pi}{\lambda} = 0.56\ cm^{-1}$

$2\pi f = 12\ sec^{-1}$

$\alpha x + \beta t + \frac{\pi}{6} = \frac{12.56\times180}{3.14} + 30 = 750^o$

$y = 7.5\ cm \sin 750^o = 3.75\ cm.$

$v = \frac{dy}{dt} = A\beta \cos\left(\alpha x + \beta t + \frac{\pi}{6}\right)$

$= 7.5 \times 12 \times \frac{\sqrt{3}}{2} = 77.94\ cm/sec.$

A-5. $\omega = 2\pi f = 4\pi\ sec^{-1}$

$K = \frac{2\pi}{\lambda} = 2\pi\ m^{-1}$

$\therefore\ y = 0.5 \cos(2\pi x + 4\pi t)$

A-6. Distance between boat = $\lambda/2 = 10$ m

$\Rightarrow \quad \lambda = 20m$

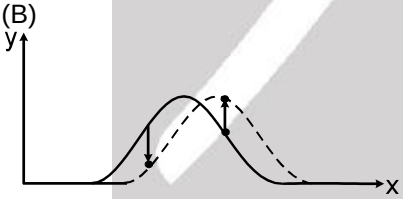
time period T = 4 sec .

$\therefore \quad V = \lambda/T = 20\ m / 4sec.= 5m/s.$

A-7. $R_A = \frac{V}{V_A},\ R_B = \frac{V}{V_B}$

as $V_A > V_B$, $R_A < R_B$

A-8. (B)



Dotted shape shows pulse position after a short time interval. Direction of the velocities are decided according to direction of displacements of the particles.

at x = 1.5 slope is +ve

at x = 2.5 slope is –ve

A-9. $v = f\lambda$

$= \frac{54}{60} \times 10 = 9\ m/sec.$

B-1. $V \propto \sqrt{T}$

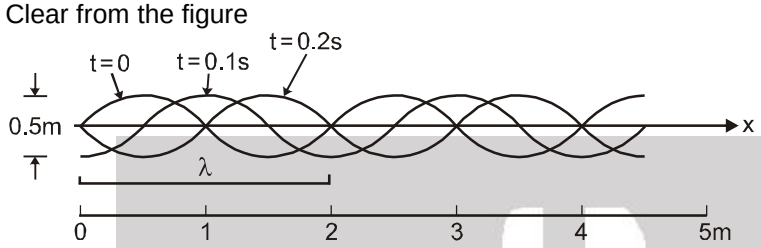
$\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{2T}{T}} = \sqrt{2}$

B-2. $V_{AB} = \sqrt{\frac{6.4g}{10\times10^{-3}}} = \sqrt{6400} = 80\ \frac{m}{sec}$

$V_{CD} = \sqrt{\frac{3.2g}{8\times10^{-3}}} = \sqrt{4000} = 20\sqrt{10}\ \frac{m}{sec}$

$V_{DE} = \sqrt{\frac{1.6g}{10\times10^{-3}}} = \sqrt{1600} = 40$ m/s

B-3. Clear from the figure



B-4. $50 = \sqrt{\frac{mg}{\mu}}$ .....(1)

$52 = \sqrt{\frac{m(g^2+a^2)^{1/2}}{\mu}}$ .....(2)

Solve (1) and (2)

$\therefore a = 4.1\ \frac{m}{sec^2}$

C-1. $p_0 = A_0^2\ \omega_0^2\ \mu v$

$\frac{p_0}{2} = A^2\ \omega^2\ \mu v$

$\therefore 2 = \frac{A_0^{\ 2}\ \omega_0^{\ 2}}{A^2\omega^2}$

$\therefore$ As, $\omega = \omega_0$ (frequency remains the same)

$\therefore A = \frac{A_0}{\sqrt{2}}$

C-2. By defination

C-3. As $\langle P\rangle = 2\pi^2 f^2 A^2 \mu v$ put values

$90 = 2 \times 10 \times f^2 \times 25 \times 10^{-4} \times 4 \times 10^{-2}\ \sqrt{\frac{100}{4\times10^{-2}}}$

$\Rightarrow$ f = 30 Hz

C-4. $P\ \alpha\ A^2\ \ \frac{P}{0.40} = \frac{4^2}{2^2}$ $\Rightarrow$ P = 1.6 watt

D-1. By defination

D-2. By definition

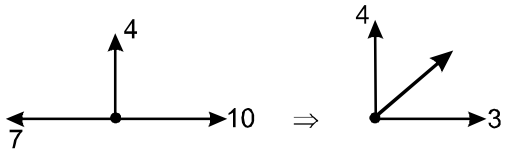
D-3. Amplitude varies between $A_1 + A_2$ to $A_1 - A_2$

D-4. Reflected wave is inverted so second string is denser so speed will decreases.

D-5. $A^2_{net} = (y_m + y_m)^2$ for $\phi = o$
$\therefore A_{net} = 2y_m \quad I \Rightarrow \quad I_{net} \alpha A^2_{net} \alpha 4y_m^2\, 4I$
where $I \Rightarrow$ Intensity of either wave

D-6. Path difference is λ between B and G.

D-7.



Resultant Amplitude = $\sqrt{3^2+4^2}$ = 5μm

E-1. As x = 0 is node $\Rightarrow$ standing wave should be y = 2a sin kx sin ωt
Now solve

E-2. $\frac{n}{2\ell}\sqrt{\frac{T}{\mu}} = 350$ and $\frac{n+1}{2\ell}\sqrt{\frac{T}{\mu}} = 420$

$\therefore \quad \frac{n}{n+1} = \frac{350}{420} \Rightarrow n = 5 \qquad \therefore \quad \frac{5\lambda}{2} = \ell \Rightarrow \lambda = \frac{2\lambda}{5}$

$\frac{v}{f} = \frac{2\ell}{5}$

$\Rightarrow \quad \frac{v}{2\ell} = \frac{f}{5}$

$\Rightarrow \quad f' = \frac{f}{5} = 70$ Hz

E-3. At $x_1 = \frac{\pi}{3K}$ and $x_2 = \frac{3\pi}{2K}$
Nodes are not formed because neither x_1 nor x_2 gives sin kx = 0

$\therefore \Delta x = x_2 - x_1 = \frac{7\pi}{6K}$

$\therefore \phi_1 = \pi$ and $\phi_2 = K\Delta x = \frac{7\pi}{6}$

$\therefore \frac{\phi_1}{\phi_2} = \frac{6}{7}$

E-4. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{A\rho}} = \sqrt{\frac{T}{\pi r^2 \rho}}$

$\frac{f_1}{f_2} = \sqrt{\frac{T_1}{T_2}\cdot\frac{\rho_2}{\rho_1}\cdot\frac{r_2^2}{r_1^2}} = \sqrt{2\times\frac{1}{2}\times\frac{1}{4}} = \frac{1}{2}$

E-5. $f = \frac{4V}{2L} \Rightarrow \lambda = \frac{v}{f} = \frac{L}{2} = 40$ cm

E-6. $\mu = \frac{m}{\ell} = \rho A$

$\therefore m_1 = \rho\pi r^2$

$m_2 = \rho 4\ \pi r^2$

$\therefore \frac{v_1}{v_2} = \frac{\sqrt{T/\mu_1}}{\sqrt{T/\mu_2}}$

Let P loops and q loops are formed respectively 1st and 2nd ωire.

$\therefore \quad \frac{p}{2\ell} V_1 = \frac{q}{2\ell} V_2 \quad \Rightarrow \quad \frac{p}{q} = \frac{1}{2}$

E-7. $y = a \sin \omega t \cos Kx$

$y = 1/2\ (2a \sin \omega t \cos Kx)$

$\therefore$ Amplitude of component wave is $a/2$

PART - III

1. Use x = 0; t = 0 for y and particle velocity $\frac{\partial y}{\partial t}$. Like for (a), y = 0 at x = 0 and t = 0. $\frac{\partial y}{\partial t} > 0$ i.e. positive therefore it matches with (R).

2. (A) Number of loops (of length $\lambda/2$) will be even or odd and node or antinode will respectively be formed at the middle.
Phase difference between two particle in same loop will be zero and that between two particles in adjacent loops will be π.
(B) and (D) Number of loops will not be integral. Hence neither a node nor an antinode will be formed in in the middle.
Phase of difference between two particle in same loop will be zero and that between two particles in adjacent loops will be π.

EXERCISE-2

PART - I

1. $\frac{\text{measure AB}}{\text{measure CD}} = \frac{\lambda}{T} = V$

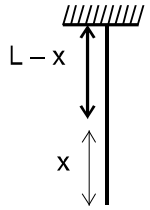
2. v is same for all the waves.
From the figure $\lambda_1 = \lambda_3 < \lambda_2$
So $\omega_1 = \omega_3 > \omega_2$

3. $V_\omega = \frac{\omega}{k} = \frac{420}{21} = 20$

$\therefore \quad V = \sqrt{\frac{T}{\mu}} = 20$

$\Rightarrow \quad T = (20)^2\ \mu = (20)^2 \times 0.2 = 80N$

4.



For the pulse

$$V = \sqrt{\frac{\mu x g}{\mu}} = \sqrt{xg} = \frac{dx}{dt}$$

$$\frac{dx}{dt} = \sqrt{xg} \Rightarrow \int_0^x \frac{dx}{\sqrt{x}} = \sqrt{g}\int_0^t dt$$

$$t = 2\sqrt{\frac{x}{g}} \qquad \text{———(1)}$$

for the particle

$$L - x = \frac{1}{2} gt^2$$

$$t = \sqrt{\frac{2(L-x)}{g}} \qquad \text{———(2)}$$

from equation (1) & (2)

$$\Rightarrow \quad \therefore x = \frac{L}{3} \text{ from the bottom}$$

5. If $T = mg = v\rho g$

$$\therefore \qquad f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = 300 \qquad \text{.....(1)}$$

Now $T' = mg - f_b = v\rho g - \frac{V}{2} g.$

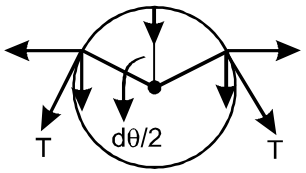
$$T' = vg\left(\frac{2\rho - 1}{2}\right)$$

$$\therefore \qquad f' = \frac{1}{2\ell}\sqrt{\frac{vg}{\mu}\frac{(2\rho-1)}{2}} \qquad \text{.....(2)}$$

$$\therefore \qquad \frac{f'}{f} = \left(\frac{2\rho-1}{2\rho}\right)^{1/2} \qquad f' = 300\left(\frac{2\rho-1}{2\rho}\right)^{1/2}$$

6.

(a)



$$2T \sin\frac{d\theta}{2} = dm.\omega^2 R$$

$$2T\frac{d\theta}{2} = \frac{m}{\ell} Rd\theta.\omega^2 R \qquad T = \frac{m\omega^2 R^2}{\ell}$$

$$\therefore \quad V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\frac{m}{\ell}}} = \omega R \qquad \therefore \qquad V = \frac{\omega L}{2\pi}$$

(b) $V_{P/R} = \frac{\omega L}{2\pi}$

$V_{P./G} - V_{R/G} = \frac{\omega L}{2\pi} \quad \Rightarrow \quad V_{P/G} - \frac{\omega L}{2\pi} = \frac{\omega L}{2\pi}$

$\therefore \quad V_{P/G} = 2\left(\frac{\omega L}{2\pi}\right) \quad V_{P/G} = \frac{\omega L}{\pi}$

(c) $V_{P/R} = \frac{\omega L}{\pi} \quad V_{P/G} - \left(-\frac{\omega L}{2\pi}\right) = \frac{\omega L}{2\pi}$

$\therefore \quad V_{P/G} = 0$

7. Satisfy the standard equation of wave

8. $\frac{n}{2\ell}\sqrt{\frac{T}{\mu}} = 384 \qquad \frac{n-1}{2\ell}\sqrt{\frac{T}{\mu}} = 288$

$\therefore \quad \frac{n}{n-1} = \frac{4}{3}$

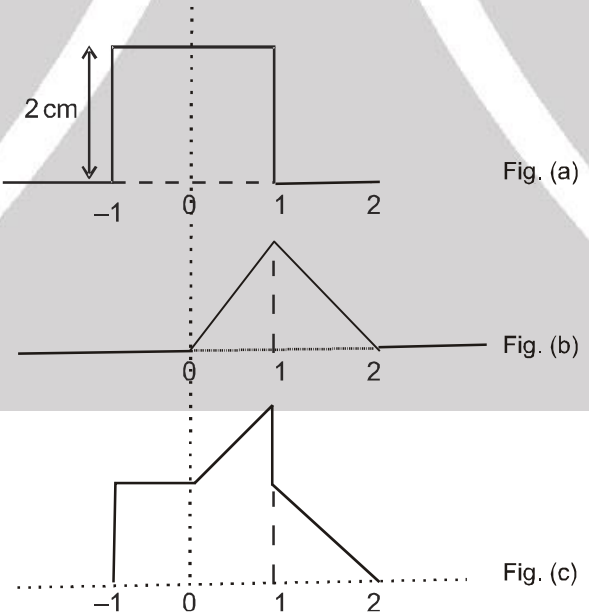
$\therefore \quad n = 4$ Now $4\left(\frac{V}{2L}\right) = 384$

Put L = 75 cm

$\therefore$ V = 144 m/sec.

9. By difination

10. At t = 2 second, the position of both pulses are separately given by fig.(a) and fig. (b); the superposition of both pulses is given by fig. (c)



11. Reflected pulse will be inverted as it is reflected by a denser medium. The wall exerts force in downward direction.

12. As $y = A \sin (Kx - \omega t + 30^o)$ for incident wave
Now for reflected wave : Energy α Amp^2
$\therefore$ $Y = 0.8\ A \sin (-Kx - \omega t + 30 + 180)$
$Y = 0.8\ A \sin (-Kx - \omega t + 210)$
$Y = -0.8\ A \sin (kx + \omega t - 210)$
$Y = -0.8\ A \sin [Kx + \omega t - 30 - 180]$
$Y = 0.8\ A \sin [180 - (Kx + \omega t - 30)]$
$Y = 0.8\ A \sin (Kx + \omega t - 30)$

13. $K = 0.025\ \pi = \dfrac{2\pi}{\lambda}$ $\qquad \lambda = \dfrac{2cm}{0.025}$

Required length $= \dfrac{\lambda}{2} = \dfrac{1}{0.025} = 40$ cm

14. $\dfrac{\omega}{K} = V_\omega$ for either component waves

15.



$n_1 = \dfrac{v}{2\ell_1}$ $\qquad n_2 = \dfrac{v}{2\ell_2}$

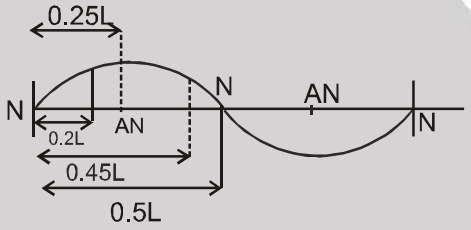
and $n = \dfrac{v}{2\ell}$ (for complete length of wire)

As $\ell = \ell_1 + \ell_2 + ...$

$$\frac{V}{2n} = \frac{V}{2n_1} + \frac{V}{2n_2} + ...$$

$$\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + ...$$

16. It is obvious that particle at 0.2 L will have larger amplitude that particle at 0.45 L, 0.5 L being the node and 0.25 being antinode.



17. $f = \dfrac{1}{2\times 40}\sqrt{\dfrac{mg}{\mu}} = 256$ .....(1)

$f = \dfrac{1}{2\times 22}\sqrt{\dfrac{mg - f_b}{\mu}} = 256$ .....(2)

$$\frac{mg - f_b}{mg} = \left(\frac{22}{40}\right)^2$$

$$\frac{v(1)g}{v\sigma g} = \frac{40^2 - 22^2}{40^2}$$

$$\sigma = \frac{40^2}{40^2 - 22^2}$$

18. $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\times 1}\sqrt{\frac{YA}{\ell}.\frac{\Delta\ell}{\rho A}} = \frac{1}{2}\sqrt{\frac{9\times10^{10}\times5\times10^{-14}}{9\times10^{3}}} = 35\ \text{Hz}$

19. As $T = \frac{YA}{20}4 \Rightarrow \nu_0 = \frac{1}{2\times24}\sqrt{\frac{YA}{20}.\frac{4}{\mu}}$ .....(1)

and $T' = \frac{YA}{20}6 \Rightarrow \nu' = \frac{1}{2\times26}\sqrt{\frac{YA}{20}.\frac{6}{\mu}}$ .....(2)

$\therefore \frac{\nu'}{\nu_0} = \frac{24}{26}\sqrt{\frac{6}{4}}$ $\therefore$ $\nu' > \nu_0$

20. $\lambda \propto v \propto \sqrt{T}$

$\therefore \frac{1}{2.2} = \sqrt{\frac{mg - \text{upthrust}}{mg}} = \sqrt{1-\frac{1}{(r.d)}}$

Where r.d = relative density r.d = 1.26

PART – II

1. $V_{max} = A\omega = 5 \Rightarrow A\ \frac{2\pi}{4} = 5$

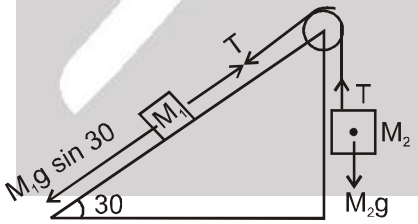
$\Rightarrow A = \frac{10}{\pi}$ cm.

2. $V = \sqrt{\frac{T}{\mu}} = 20$ m/s

s = Vt

t = 0.03 sec.

3.



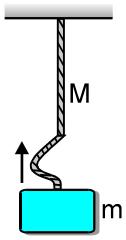
$T = M_2g = M_1g \sin 30 \Rightarrow M_1 = 2M_2$

Now $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{M_2g}{9.8\times10^{-3}}}$

$100 = \sqrt{\frac{M_2}{10^{-3}}} \Rightarrow M_2 = 10$kg

$\therefore$ $M_1 = 20$ kg

4.



$f = \left(\frac{v}{\lambda}\right)$ at lowest point

$f = \frac{1}{\lambda}\sqrt{\frac{mg}{\mu}}$

$f^1 = \frac{v^1}{\lambda^1}$ at highest point

$f^1 = \frac{1}{\lambda^1}\sqrt{\frac{(M+m)g}{\mu}}$

$f^1 = f$: no any change in source frq.

$\frac{1}{\lambda}\sqrt{\frac{mg}{\mu}} = \frac{1}{\lambda}\sqrt{\frac{(M+m)g}{\mu}} \Rightarrow \left[\lambda^1 = \lambda\sqrt{\frac{M+m}{m}}\right]$

5. (a) $\mu = Kx = \frac{dM}{dx}$

$$\int_O^M dM = \int_O^L Kx\ dx$$

$$M = \frac{KL^2}{2}$$

and $K = \frac{2M}{L^2}$

(b) $V = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{F}{Kx}} = \frac{dx}{dt}\int_0^L \sqrt{x}\ dx = \sqrt{\frac{F}{K}}\int_o^t dt$

$\therefore \quad t = \sqrt{\frac{4L^3}{9}\cdot\frac{K}{F}} = \sqrt{\frac{4L^3}{9F}\cdot\frac{2\ M}{L^2}} = \sqrt{\frac{8ML}{9\ F}}$

6. $2f = 6, \quad f = 3, \quad T = \frac{1}{3}$ sec, $\qquad v = 3$ m/sec

$3 = \frac{\lambda}{T} \Rightarrow \lambda = 1$m

$x = vt$

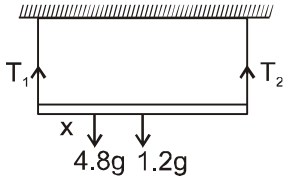
$3 = 3t \Rightarrow t = 1$

total time $= t + \frac{3T}{4}$

$= 1 + \frac{3}{4}\left(\frac{1}{3}\right) = \frac{5}{4} = 1.25$ sec.

7. $\frac{\lambda_1}{2} = \ell \Rightarrow \lambda_1 = 2\ell \qquad \lambda_2 = \ell$

$\therefore \frac{\lambda_2}{\lambda_1} = \frac{1}{2} \Rightarrow \frac{\lambda_1}{\lambda_2} = 2 \Rightarrow \frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}} = 2$



$T_1 = 4T_2$ ——(1)

$T_1 + T_2 = 60$

$\therefore$ $T_1 = 48$ N

$T_2 = 12$ N

Now taking moments about right end

$48 \times 0.4 = 48 \times (0.4 - x) + 0.2 \times 12$

Put T_1 & T_2

$\therefore$ $x = 5$ cm $\therefore$ i. e 5 cm from left. 5 cm

8. For third overtone $\frac{4\lambda}{2} = \ell$

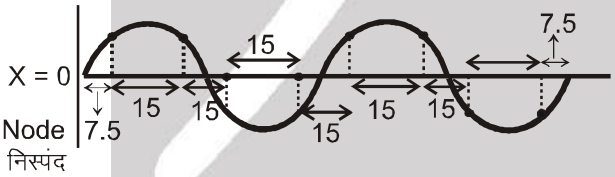
$\Rightarrow 2\lambda = \ell \Rightarrow \lambda = \frac{\ell}{2}$

As $x = 0$ is a node

$\therefore A_{s.\omega.} = A \sin \frac{2\pi}{\lambda} x$

$= a \sin\left(\frac{2\pi}{\frac{\lambda}{2}}\frac{1}{3}\right) = a \frac{\sqrt{3}}{2}$

9.



(a) $\therefore X = 7.5 \qquad \lambda = 60$

$\therefore A_{S\Omega} = 3.5 = A \sin Kx$

$3.5 = A \sin\left(\frac{2\pi}{60}7.5\right) \Rightarrow A = 3.5\ \sqrt{2}$ mm

$\therefore A = 3.5\ \sqrt{2}$ mm

(b) 3rd overtone $\Rightarrow$ ωhich covers 120 cm length of the string

10. Energy $E = \int_0^{\ell} \frac{1}{2} dm v^2 = \int_0^{\ell} \frac{1}{2} dm A_x^{\ 2} \quad \omega^2 = \int_0^{\ell} \frac{1}{2}\left(\frac{m}{\ell}\right) dx \quad A^2 \sin^2 kx \quad \omega^2$

$= \int_0^{\ell} \frac{1}{2}\left(\frac{m}{\ell}\right) \quad A^2\omega^2 \sin^2 kx \quad dx = \frac{1}{4} m A^2 \omega^2$

As $\omega = 2\pi f = 2\pi \frac{v}{2\ell} = \frac{\pi}{\ell}\sqrt{\frac{T}{\mu}} = \frac{\pi}{\ell}\sqrt{\frac{T\ell}{m}}$

$\therefore$ Energy $= \frac{1}{4}\ m\ a^2 \frac{\pi^2}{\ell^2} \frac{T\ell}{m} = \frac{a^2\pi^2 T}{4\ell}$

11. $A_i = 5$ A

$A_r = 3$ A

As it is the care of partially standing wave

$\therefore \quad A_{NODE_{\text{निस्पंद}}} = A_i - A_r$

$= 2$ A

and $\quad A_{ANTINODE_{\text{प्रस्पंद}}} = A_i + A_r = 8$ A

12. Use ; $n = \dfrac{2}{2\ell}\sqrt{\dfrac{T}{\mu}}$

13. L = 1 m

M = 0.04 kg

$\mu = \dfrac{M}{L} = 0.04$ kg/m

$\therefore V = \sqrt{\dfrac{T}{\mu}} = \sqrt{\dfrac{400}{0.04}} = 100$ m/sec

For fundamental frequency

$L = \dfrac{\lambda}{4} \Rightarrow \lambda = 4L = 4$ m

$\therefore f_1 = \dfrac{100}{4} = 25$ Hz

(i) $\dfrac{\lambda_1}{4} = L = 1$

$\lambda_1 = 4$ m

(ii) $\dfrac{3\lambda_2}{4} = L = 1$

$\lambda_2 = \dfrac{4}{3}$ m

(iii) $\dfrac{5\lambda_3}{4} = L$

$\lambda_3 = \dfrac{4}{5}$ m

14. $\ell = 90\text{cm} = 0.9\text{m}$

$\mu = \dfrac{44\times10^{-3}}{0.9} = \left(\dfrac{44\times10^{-3}}{90\times10^{-2}}\right)$

$f = \dfrac{4v}{2L} = \left(\dfrac{2v}{L}\right) \Rightarrow \quad 60 = \left(\dfrac{2v}{\ell}\right)$

$v = \dfrac{60\ell}{2} = 30l = 30 \times 0.9 = 27$

$\sqrt{\dfrac{T}{\mu}} = 27$

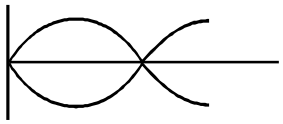
$T = (27 \times 27 \times \mu) = 27 \times 27 \times \dfrac{44\times10^{-3}}{90\times10^{-2}}$

(T = 36 N)

15. $\frac{n}{2\ell}\sqrt{\frac{T}{\mu}} = 90$

$\frac{n+1}{2\ell}\sqrt{\frac{T}{\mu}} = 150 \quad \frac{n+2}{2\ell}\sqrt{\frac{T}{\mu}} = 210$

$\therefore \frac{n}{n+1} = \frac{90}{150} \Rightarrow n = 1.5$



$\therefore$

$\therefore \frac{n}{2\ell}\sqrt{\frac{T}{\mu}} = 90 = \frac{n}{2\ell}V$

$\frac{V}{2\ell} = \frac{90}{n} = \frac{90}{1.5} = 60$

$\therefore \frac{V}{4\ell} = 30$

(b) $\frac{3V}{4L} = 90$ and और $\frac{5V}{4\ell} = 150$

and $\frac{7V}{4\ell} = 210$

(c) overtones are 1st, 2nd, 3rd

(d) $\frac{v}{4\times 0.8} = 30 \Rightarrow v = 96$ m/sec.

16. $f = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{YA\alpha\Delta T}{\mu}}$

Put values f =11

17. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{4.5\times 10^7}{0.05}}$

$\frac{n}{2\ell}\sqrt{\frac{4.5\times 10^7}{0.05}} = 420$ ———(1)

$\frac{n+1}{2\ell}\sqrt{\frac{4.5\times 10^7}{0.05}} = 490$ ———(2)

From equation (1) & (2) $\Rightarrow \frac{n}{n+1} = \frac{6}{7} \Rightarrow n = 6$

Put n in (1) $\therefore \frac{6}{2\ell}\ 3\times 10^2 = 420$

$\therefore \frac{6}{2\ell}\ 3\times 10^2 = 420$

$\ell = \frac{30000}{140} \qquad \ell = \frac{1500}{7} = 214$ cm

18. Initially

$T = mg$

$f = 10\, f_0 = 10\, \frac{V}{2\ell}$

$f = \frac{10}{2\ell}\sqrt{\frac{Mg}{\mu}}$ .....(1)

Finally ; $\nu = \frac{11}{2\ell}\sqrt{\frac{mg - f_b}{\mu}}$ .....(2)

$1 = 2 \Rightarrow$ $100\, mg = 121\,(mg - f_b)$

$100\, m = 121m - 121\, V\rho_w$

$100\, V\sigma = 121\, V\sigma - 121\, V\rho_w$ (σ is the density of block)

$21/\sigma = 121/\rho_w$ ($\rho_w = 1$)

$\sigma = \frac{121}{21} = 5.8\ \frac{gm}{cm^3}$

19. Let number of loops in steel are P and in aluminium number of loops are Q.

$\therefore \frac{P}{2\ell_s}\sqrt{\frac{T}{\mu_s}} = \frac{Q}{2\ell_{Al}}\sqrt{\frac{T}{\mu_{Al}}}$

$\therefore \frac{P}{Q} = \frac{4}{3}$ .....(1)

For minimum frequency

$f = \frac{4}{2 \times \frac{80}{100}}\sqrt{\frac{40}{10^{-6} \times 7.8 \times 10^3}}$

$f = 180$ Hz

20. $f_1 = \frac{1}{2\ell_1}\sqrt{\frac{T\ell_1}{m}}$ $f_2 = \frac{1}{2\ell_2}\sqrt{\frac{T\ell_2}{m}}$

$\therefore \frac{f_1}{f_2} = \sqrt{\frac{\ell_2}{\ell_1}} = \sqrt{\frac{\ell_1(1 - 20\alpha)}{\ell_1}}$

$\frac{f_1}{f_2} = (1 - 10\alpha)$ (By Binomial theorem, assuming temperature change to be small)

$\alpha = \frac{f_2 - f_1}{10 f_2} = 10^{-4}\ ^oC^{-1}$

PART - III

1*. Standard equation

2*. Compare with $y = a \sin(\omega t + Kx)$

3*. Compare with $y = A \sin(\omega t - Kx)$

4*. Satisfy the standard equation of wave

5*. Comparing with $y = A\cos(\omega t - kx)$

$\omega = 500\ s^{-1}$, $k = 0.025\ m^{-1}$,

$v = \dfrac{500}{0.025} = 2 \times 10^4$ m/s

$\lambda = \dfrac{500}{0.025} = 80\,\pi$ m

$y = A\cos(\omega t - kx)$

6*. $V_{max} = A\omega = \dfrac{v}{10} \Rightarrow \quad 10^{-3}\,2\pi f = \dfrac{10}{10}$

$\Rightarrow \quad f = \dfrac{1000}{2\pi}$ Hz

Now $V = f\lambda$ find λ

7*. $V \alpha \sqrt{T}$

$f \alpha \sqrt{T}$

$$\frac{f}{f+15} = \left(\frac{T}{T+\frac{21T}{100}}\right)^{1/2} = \frac{10}{11}$$

$11f = 10f + 150$

$f = 150$ Hz

8. $y_1 = A\sin(2\pi f + t - kx)$

$2\pi f = \omega$

$y_2 = 3A\sin(2\pi f + t + kx)$

due to superposition $(y = y_1 + y_2)$

$\Rightarrow y = (A\sin(\omega t - kx) + 3A\sin(\omega t + kx))$

$y = (4A\cos kx \sin \omega t - 2A \sin kx \cos \omega t)$

So at the point on the path average displacement is zero

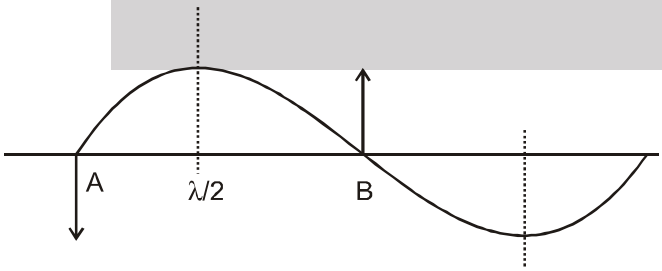
$A^1 \sin\phi = 2A \sin kx$

$A^1 \cos\phi = 4A \cos kx$

$A^1 = \sqrt{4A^2\sin^2 kx + 16A^2\cos^2 kx}$

So A^1 change with x (position)

9*.



A and B point will move in opposite direction displacement level of A and B will be equal

11*. Use principle of superposition

12*. Compare with $y = A\cos\left(\omega t + \frac{\pi}{2}\right)\cos(kx)$

$y = A\cos\left(\omega t + \frac{\pi}{2}\right)\cos(kx)$

i.e. $y = A\sin\omega t\cos Kx$

$\omega = 50\pi$, $k = 10\pi$, $v = 5$ m/s

$\frac{2\pi}{\lambda} = 10\pi$

$\lambda = \frac{1}{5} = 0.2$ m

Position of antinode $= 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, 2\lambda$

$= 0, 0.1, 0.2, 0.3$

position of node $= 0.05, 0.15, 0.25,$

13*. Compare with $y = A\sin Kx\cos\omega t$

$y = A\sin Kx\cos\omega t$ से तुलना करें

14*. $\ell = 1$ m $\quad \varepsilon = 10\sin(80\pi t - 4\pi x)$

For superimposed second wave is

$\varepsilon_2 = 10\sin(80\pi t + 4\pi x)$

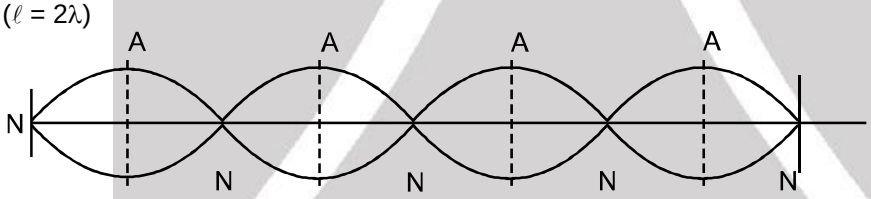
Amplitude of stationary wave = $2A = 2 \times 10 = 20$m

$K = 4\pi = \frac{2\pi}{\lambda}$

$\lambda = \frac{2\pi}{4\lambda} = \frac{1}{2} = 0.5$ m

$\ell = 1$m

$(\ell = 2\lambda)$



Total (N = 5)

15.



v_{max} at $y = 0$

So K_{mat} at $y = 0$

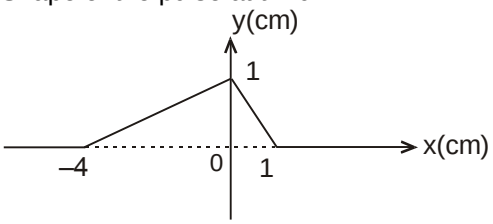
$\Rightarrow$ C

So max^m P.E. at $y = 0$

$\Rightarrow$ D

PART – IV

1. Shape of the pulse at t = 0



That is a triangular pulse
Area of the pulse

$= \frac{1}{2}\ [(4 \times 1) + (1 \times 1)]$

$= \frac{1}{2}\ cm^2$

2. $v = \sqrt{\frac{T}{\mu}} = 10$ m/s

Solution of the wave equation that gives displacement of any piece of the string at any time

$$y = f(x,\ t) = \begin{cases} \frac{(x - vt)}{4} + 1 & \text{for} \quad vt - 4 < x \le vt \\ -(x - vt) + 1 & \text{for} \quad vt < x < vt + 1 \\ 0 & \text{otherwise} \end{cases}$$

$$y = f(x,\ t) = \begin{cases} \frac{(x - vt)}{4} + 1 & vt - 4 < x \le vt \quad \text{के लिए} \\ -(x - vt) + 1 & vt < x < vt + 1 \quad \text{के लिए} \\ 0 & \text{अन्य था} \end{cases}$$

Using $v = 1000$ cm/s, $t = 0.01$ s
$vt = 10$ cm
as $(vt - 4) < (x = 7\ cm) < vt$

$y = \frac{1}{4}\ (7 - 10) + 1 = \frac{1}{4}$ cm = 0.25 cm

3. Transverse velocity $= \frac{\partial y}{\partial t}$

at t = 0.015 s, vt = 15 cm
as for x = 13 cm $(vt - 4) < x < vt$

$= \frac{\partial y}{\partial t}$

t = 0.015 s, vt = 15 cm
x = 13 cm $(vt - 4) < x < vt$

therefore

$\frac{\partial y}{\partial t} = -\frac{v}{4} = -250$ cm/s

4-6 f = 120 $A = \frac{1}{2} \times 10^{-2}$

$V_\omega = \sqrt{\frac{T}{\mu}} = 100$ and

$\mu = 90 \times 10^{-3}$ Kg met

$V_{P_{max}} = A\omega = \frac{1}{2} \times 10^{-2} \times 2\pi \,.\, 120 = 1.2\,\pi$

$(T_y)_{max} = (T \sin\theta)_{max} = (T \tan\theta)_{max}$

$= T\left(\frac{\delta_y}{\delta_x}\right)_{max} = T\,\frac{V_{P_{max}}}{V_w}$

$= \frac{900 \times 1.2\pi}{100} = 10.8\,\pi$ N

$P_{max} = 4\pi^2 f^2 A^2 \mu v = 12.96\pi^2$ watt

$P_{min} = 0$

When minimum power transfer occurs displacement is $= \frac{1}{2}$ cm.

7-9 Medium of string BC is denser. These will be π phase difference between reflected and transmitted wave.

$$\frac{\lambda_r}{\lambda_t} = \frac{v_1}{v_2} = \sqrt{\frac{\mu_2}{\mu_1}}$$

Now $\frac{\mu_1}{\mu_2} = \frac{4}{9}$ so $\frac{\lambda_r}{\lambda_t} = \frac{3}{2}$

$$\frac{A_r}{A_t} = \frac{\frac{v_2 - v_1}{v_1 + v_2} Ai}{\frac{2v_2}{v_1 + v_2}.Ai} = \frac{v_2 - v_1}{2v_2} = \frac{1}{4}$$

EXERCISE-3

PART - I

1. Maximum particle velocity $v_{max} = \omega A = 3$ m/s

Maximum particle acceleration $a_{max} = \omega^2 A = 90$ m/s^2

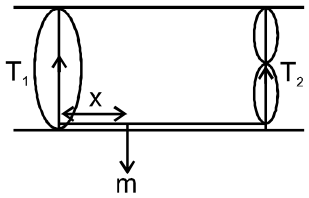
$a_{max} = v_{max}\,\omega = \omega \times 3 = 90$ m/s^2 $\Rightarrow$ $\omega = 30$ s^{-1} $\Rightarrow$ $A = \frac{3}{\omega} = \frac{3}{30} = 0.1$ m

$k = \frac{\omega}{v} = \frac{30}{20} = \frac{3}{2}$ [where v is velocity of wave]

Equation of wave in string $y = 0.1 \sin\left(30\ \ t\ \ \pm\frac{3}{2}x + \phi\right)$ [where ϕ is initial phase]

2. $\frac{\lambda_1}{2} = \ell \Rightarrow \lambda_1 = 2\ell$

$\lambda_2 = \ell \quad \Rightarrow \therefore \frac{\lambda_1}{\lambda_2} = 2$



$\frac{v_1/f}{v_2/f} = 2$

$\frac{v_1}{v_2} = 2 = \sqrt{\frac{T_1/\mu}{T_2/\mu}} \Rightarrow \frac{T_1}{T_2} = 4$ ——(1)

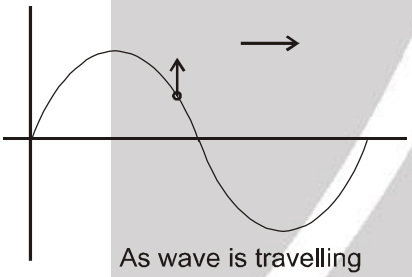
Now moment about P : $T_1 x = T_2 (\ell - x)$

$\ell - x = 4x \qquad x = \ell/5$

3. $V = \omega \sqrt{A^2 - y^2}$

$V_P = 2\pi f \sqrt{A^2 - y^2}$

$V_P = 2\pi \left(\frac{V}{\lambda}\right) \sqrt{A^2 - y^2} = \frac{2\pi}{0.5} \times 0.1 \sqrt{(0.1)^2 - (0.05)^2}$



As wave is travelling in +ve direction

$V_P = \frac{\sqrt{3}\pi}{50}$ j m/s **Ans. (A)**

4. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{0.5}{10^{-3}/0.2}} = 10$ m/sec.

$v = f\lambda$

$10 = (100)\lambda \quad \Rightarrow \quad \lambda = 0.1$ m $= 10$ cm

Distance between two successive nodes = $\frac{\lambda}{2} = 5$ cm

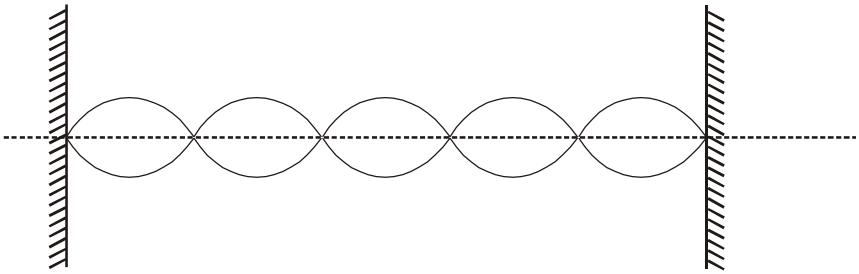
Ans. 5

5. $A_{eq} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$

$A_{eq} = \sqrt{4^2 + 3^2 + 2(4)(3)\cos\frac{\pi}{2}}$

$A_{eq} = 5.$

6. (A) There are 5 complete loops.
Total number of nodes = 6



(B) $\omega = 628\ sec^{-1}$

$$k = 62.8\ m^{-1} = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{1}{10}$$

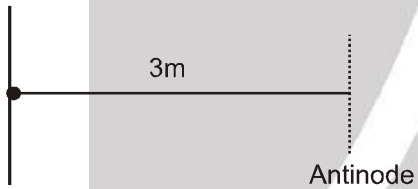
$$v_w = \frac{\omega}{k} = \frac{628}{62.8} = 10\ ms^{-1}$$

$$L = \frac{5\lambda}{2} = 0.25$$

(C) 2A = 0.01 = maximum amplitude of antinode

(D) $f = \frac{v}{2\ell} = \frac{10}{2\times 0.25} = 20$ Hz.

7. V = 100 m/s



Possible modes of vibration

$$\ell = (2n+1)\frac{\lambda}{4}$$

$$\lambda = \frac{12}{(2n+1)}\ m$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{12/(2n+1)} = \frac{(2n+1)\pi}{6}$$

$$\omega = vk = 100\,(2n+1)\frac{\pi}{6} = \frac{(2n+1)\ 50\pi}{3}$$

if $n = 0 \quad k = \frac{\pi}{6} \quad \omega = \frac{50\ \pi}{3}$

$n = 1 \quad k = \frac{5\ \pi}{6} \quad \omega = \frac{250\ \pi}{3}$

$n = 7 \quad k = \frac{5\ \pi}{2} \quad \omega = 250\ \pi$

8.* $V = \sqrt{\frac{T}{\mu}}$, so speed at any position will be same for both pulses, therefore time taken by both pulses will be same.

$\lambda f = v \Rightarrow \lambda = \frac{V}{f} \Rightarrow \lambda \propto V$, since when pulse 1 reaches at A, speed decreases therefore λ decreases.

At mid point, magnitude of velocity is same, but direction will be opposite. Hence velocity will be in opposite direction.

PART – II

1. $(n + 1)\ \frac{v}{2\ell} = 420$ .....(1)

$\frac{nv}{2\ell} = 315$ .....(2)

(1) – (2) $\frac{V}{\mu} = 105$ Hz $f_{min} = = 105$ Hz

2. $\frac{2\pi}{\lambda}\pi = \alpha$ $\alpha = \frac{2\pi}{0.08} = 25\pi$ $\frac{2\pi}{T} = \beta$ $\beta = \frac{2\pi}{2} = \pi$

3. By equation

$f = \frac{1}{0.04}$ and $\lambda = 0.5$

$\Rightarrow$ $V = \frac{1}{0.04} \times 0.5 = \frac{25}{2}$

by $V = \sqrt{\frac{T}{\mu}}$ $\Rightarrow$ $\left(\frac{25}{2}\right)^2 = \frac{T}{0.04} \Rightarrow T = \frac{625}{4} \times 0.04$

T = 6.25 N

4. $y(x,t) = e^{-[\sqrt{a}x+\sqrt{b}t]^2}$

It is transverse type $y(x,t) = e^{-(ax+bt)^2}$

Speed $v = \frac{\sqrt{b}}{\sqrt{a}}$

and wave is moving along –x direction.

$y(x,t) = e^{-[\sqrt{a}x+\sqrt{b}t]^2}$

5. $Y = A\sin(\omega t - kx) + A\sin(\omega t + kx)$

$Y = 2A\sin\omega t\cos kx$ standing wave

For nodes $\cos kx = 0$

$\frac{2\pi}{\lambda}.x = (2n + 1)\ \frac{\pi}{2}$

$\therefore$ $x = \frac{(2n+1)\lambda}{4}$, n = 0,1,2,3,.....

6. $f = \frac{v}{2\ell} = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}} = \frac{1}{2\ell}\sqrt{\frac{T}{Ad}}$

Also $Y = \frac{T\ell}{A\Delta\ell} \Rightarrow \frac{T}{A} = \frac{Y\Delta\ell}{\ell}$

$\Rightarrow \quad f = \frac{1}{2\ell}\sqrt{\frac{y\Delta\ell}{\ell d}}$

$\ell = 1.5\text{m}$, $\frac{\Delta\ell}{\ell} = 0.01$, $d = 7.7 \times 10^3$ kg/m^3

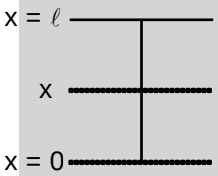
$y = 2.2 \times 10^{11}$ N/m^2

After solving

$f = \sqrt{\frac{2}{7}} \times \frac{10^3}{3}$ Hz.

$f \approx 178.2$ Hz. **Ans. (2)**

7. Let mass per unit length be λ.



$T = \lambda g x \qquad v = \sqrt{\frac{T}{\lambda}} = \sqrt{gx}$

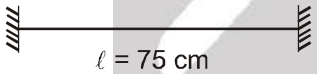
$v^2 = gx$,

$a = \frac{vdv}{dx} = \frac{g}{2}$

$\ell = \frac{1}{2}\frac{g}{2}t^2 \Rightarrow t = \sqrt{\frac{4\ell}{g}} = 2\sqrt{2}$ sec

HIGH LEVEL PROBLEMS (HLP)
SUBJECTIVE QUESTIONS

1.



$f_1 = 420 = \frac{nv}{2L}$

$f_2 = 315 = \frac{mv}{2L}$

$\frac{f_1}{f_2} = \frac{420}{315} = \frac{n}{m}$ (here, n and m are integers)

So $n = 4$, $m = 3$

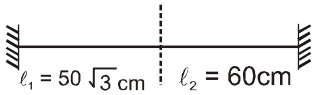
No any resonant frequency so

$420 = \frac{4v}{2L}$

$f_2 = \frac{v}{2L} = \left(\frac{420}{4}\right) = 105\ H_2$ (Lowest resonant frequency.)

$V = (105)\ (2L) = 105 \times 2 \times 0.75 = 105 \times \frac{3}{2} = 157.5$ m/sec.

2.



Steel Aluminium

T = 104

$\sigma_1 = 7.8$ gm / cm^3 $\sigma_2 = 2.6$ gm /cm^3

$A_1 = A_2 = 1$ mm^2

$$V_1 = \sqrt{\frac{T}{\sigma_1 A_1}} = \sqrt{\frac{104}{\frac{7.8\times10^{-3}}{10^{-6}}\times1\times10^{-6}}}$$

$$V_1 = \left(\frac{200}{\sqrt{3}}\right) \text{ m/ sec.}$$

$$v_2 \sqrt{\frac{T}{\sigma_2 A_2}} = \sqrt{\frac{104}{2.6\times10^{-3}}}$$

$= 2 \times 10^2$ m/sec.

for lowest freq.

$f_1 = f_2$

$$\frac{n_1 V_1}{2L_1} = \frac{n_2 V_2}{2L_2}$$

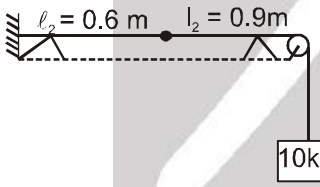
$$\frac{n_1\times200}{2\times\sqrt{3}\times50\sqrt{3}} = \frac{n_2\times200}{2\times60\times1}$$

$$\Rightarrow \left[\frac{n_1}{n_2} = \left(\frac{5}{2}\right)\right]$$

So lowest freq.

$$f_2 = \frac{n_2 V}{2L} = \frac{2\times200}{2\times0.6} = \frac{1000}{3} \text{ Hz}$$

3.



$A = 1 \times 10^{-6}$ m, $T = 10g = 100$N

$\sigma_1 = 2.6 \times 10^3$kg/m^2 $\sigma_2 = 1.04 \times 10^4$ kg /m^3

$$v_1 = \sqrt{\frac{100}{1\times10^{-6}\times2.6\times10^3}} = \frac{10^3}{\sqrt{26}} \text{m/sec}$$

$$v_2 = \sqrt{\frac{100}{1\times10^{-6}\times1.04\times10^4}} = \frac{10^3}{2\sqrt{26}}$$

$f_1 = f_2$

$$\frac{n_1\times10^3}{2\times0.6\sqrt{26}} = \frac{n_2\times10^3}{2\times2\times0.9\sqrt{26}} \left[\frac{n_1}{n_2} = \frac{1}{3}\right]$$

$$\text{Lowest frequency} = \frac{n_1 v_1}{2L_1} = \frac{1\times10^3}{\sqrt{26}\times2\times0.6}$$

$$= \frac{10^4}{12\sqrt{26}} = 162 \text{ vibration/sec.}$$

$\frac{n_1}{n_2} = \left(\frac{1}{3}\right)$

One side 1 loop and other side 3 loop



So excluding the two nodes at ends total nodes is 3

4. $f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

If length is constant

$\frac{f'}{f} = \left(\frac{T'}{T}\right)^{\frac{1}{2}}$

$f' = f + 6$, $T' = T + 0.44\ T = 1.44\ T$

$\frac{f+6}{f} = \sqrt{\frac{1.44T}{T}}$

$f = 30Hz$

If T is constant

$\frac{f''}{f} = \frac{\ell}{1.20\ell}$

$f'' = \frac{f}{1.2} = \frac{300}{12}$

$f'' = 25Hz$

$\Delta f = f'' - f = 25 - 30 = -5Hz$

So fundamental freq. Will decrease by 5Hz

5.



If temperature decreases,s tension in wire increases and v increases

$v_1 = \sqrt{\frac{F}{\mu}}$

$F = \mu v_1^2$

$\Delta F = YA\ \alpha\ \Delta T$

$v_2 = \sqrt{\frac{F+\Delta F}{\mu}} = \sqrt{\frac{F + YA\ \alpha\ \Delta T}{\mu}}$

$v_2^2 = v_1^2 + \frac{YA\ \alpha\ \Delta T}{\mu}$

$\frac{YA\ \alpha\ \Delta T}{(m/\ell)} = v_2^2 - v_1^2$

$\frac{Y\ \alpha\ \Delta T}{\rho} = v_2^2 - v_1^2 \qquad \alpha = \frac{\rho\ (v_2^2 - v_1^2)}{Y\ \Delta T}$

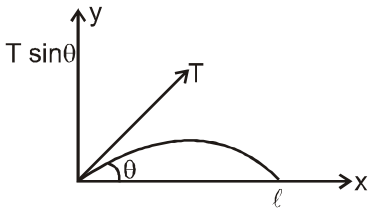
6. $V_1 = \sqrt{\frac{10}{4\times10^{-3}}}$

$= 50$ m/sec

$V_2 = \sqrt{\frac{10}{16\times10^{-3}}}$

$= \sqrt{\frac{10^4}{16}} = \frac{100}{4} = 25$ m/sec

7.



$F_y = T \sin\theta = T \tan\theta$

$\simeq T\frac{\partial y}{\partial x}$

$y = A \sin(\omega t - kx)$

$= 0.02 \sin\left(\omega t - \frac{2\pi}{0.50 \times 2}x\right)$

$\frac{\partial y}{\partial x} = -0.02 \times 2\pi \cos\left(\omega t - 2\pi x\right)$

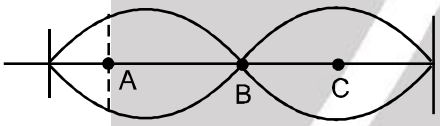
At x = 0

$\frac{\partial y}{\partial x} = -0.02 \times 2\pi \cos\omega t$

$\left[\frac{\partial y}{\partial x}\right]_{max} = 0.02 \times 2\pi$

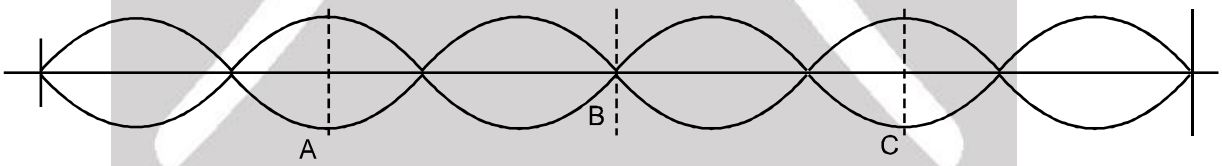
$F_y = \left[T \frac{\partial y}{\partial x}\right] = 0.02 \times 2\pi \times 10 = \frac{2\pi}{5}$ N

8.



(n = 2)

$f_1 = \frac{2v}{2L}$ ———(i)



n = 6

$f_2 = \frac{6v}{2L}$ ———(i)

$\frac{f_2}{f_1} = \frac{6v/2L}{\frac{2V}{2L}} = 3$

$f_2 = 3f_1 = 3 \times 100 = (300\ Hz)$

9. $f = \frac{1}{2L}\sqrt{\frac{F}{\mu}}$

as F and μ are fixed

$\frac{f_1}{f_2} = \frac{L_2}{L_1}$ $\quad L_2 = \frac{f_1}{f_2} L_1 = \frac{90}{135} \times 180cm = 120\ cm$

Thus the string should be pressed at 120 cm from an end

10. $\ell = 90\text{ cm} = 0.9\text{m}$

$\mu = \frac{4}{90}\text{ gm/cm}$

$= \frac{4}{900}\text{ Kg/m}$

$f = 125$

$f = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

$125 = \frac{1}{2\times 0.9}\sqrt{\frac{T\times 900}{4}}$

$T = 225\text{ N}$

11. $n_1 = \frac{1}{2L_1}\sqrt{\frac{T}{\mu}}$

$n_2 = \frac{1}{2L_2}\sqrt{\frac{T}{\mu}}$

$n_3 = \frac{1}{2L_3}\sqrt{\frac{T}{\mu}}$

$n_1 : n_2 : n_3 : 1 : 2 : 3$

$n_2 = 2n_1$

$n_3 = 3n_1$

$L_2 = \frac{n_1}{n_2}\ L_1 = \left(\frac{L_1}{2}\right)$

$L_1 + L_2 + L_3 = 1.21$

$L_1 + \frac{L_1}{2} + \frac{L_1}{3} = 1.21 \qquad L_1\left(\frac{6+3+2}{6}\right) = 1.21$

$L_1\ \left(\frac{11}{6}\right) = 1.21$

$L_1 = \frac{6\times 121}{11\times 100} = \frac{6\times 11}{100} = 0.66\text{m}$

$L_2 = \frac{L_1}{2} = \frac{0.66}{2} = 0.33\text{m}$

$L_3 = \frac{L_1}{3} = \frac{0.66}{3} = 0.22\text{m}$

12. $y_p = 2\sin(100\pi t + 30^0)$

$y = 2\sin\left[100\pi(t - \frac{25}{50}) + 30^0\right] = 2\sin(100\pi t - 50\pi + 30°)$

$z = 3\sin\left[100\pi\left(t - \frac{75}{50}\right) + 60^0\right]$

$= 3\sin\left(100\pi t - 150\pi + 60^0\right) = 3\sin(100\pi t + 60°)$

$\vec{r}_{(\text{in mm ग})} = x\hat{i} + y\hat{j} + z\hat{k}$

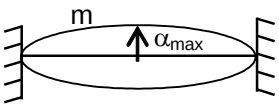
$= 25000\hat{i} + 2\sin(100\pi + 30^0)\hat{j} + 3\sin(100\pi t + 60^0)\hat{k}$

$\vec{v} = \frac{d\vec{r}}{dt} = 2\times 100\pi\cos(100\pi t + 30°)\hat{j} + 3\times 100\pi\cos(100\pi t + 60°)\hat{k}$

$\vec{v}(\text{in m/s}) = 0.2\pi\cos(100\pi t + 30°)\hat{j} + 0.3\pi\cos(100\pi t + 60°)\hat{k}$

Phase difference at time 't' = 30° constant always after they meet at 'P'.

13.



equation of standing wave :

$y = \alpha_{max} \sin kx \cos \omega t$

$V_y = \frac{dy}{dt} = -a_{max} \sin kx \sin \omega t$

$Vy = \frac{dy}{dt} = -\alpha_{max} \omega \sin kx \sin \omega t$

(a) For maximum velocity $\sin \omega t = 1$

then $V_y = -\alpha_{max} \omega \sin kx$

Then kinetic energy :

$$dk = \frac{1}{2}(dm)Vy^2$$

$$dk = \frac{1}{2}(\mu dx)\alpha_{max}^2 \omega^2 \sin^2 kx$$

$$\int dk = \frac{1}{2}\mu\alpha_{max}^2 \omega^2 \int_0^{\lambda/2} \sin^2 kx dx$$

$$k = \frac{1}{2}\mu\alpha_{max}^2 \omega^2 \left(\frac{1}{2}\right)$$

$$K = \frac{1}{4}\mu\alpha_{max}^2 \omega^2$$

(b) Average kinetic energy for strip of length dx :

$<dk> = < \frac{1}{2} dmv^2 > = < \frac{1}{2} (\mu dx) (\alpha_{max} \omega \sin kx \sin \omega t)^2 >$

$<dk> = \frac{1}{2} \mu\alpha^2_{max} \omega^2 \sin^2 kx dx \int_0^{2\pi/\omega} \sin^2 \omega t = \frac{1}{4}\alpha_{max}^2 \omega^2 \sin^2 kx dx$

Averge ower all mole cubes :

$$<k> = \frac{1}{4}\mu\alpha_{max}^2 \omega^2 < \sin^2 kx >$$

$$<k> = \frac{1}{8}\mu\alpha_{max}^2 \omega^2 \qquad \text{Ans.}$$

From (i) and (ii) :

$$dE = dU + ak$$

$$dE = \frac{1}{2} \mu a^2 \omega^2 [\sin^2 kx \sin^2 \omega t + \cos^2 kx \cos^2 \omega t] dx \quad \ldots(iii)$$

Average value of dE at (dx) element.

$$<dE> = \frac{1}{2} \mu a^2 \omega^2 \left[\sin^2 kx < \sin^2 \omega t > + \cos^2 kx < \cos^2 \omega t >\right] dx$$

$$< dE > = \frac{1}{4} \mu a^2 \omega^2 [\sin^2 kx + \cos^2 kx] dx$$

$$< E > = \frac{1}{4}\mu a^2 \omega^2 \int_0^{\lambda/2} dx = \frac{1}{4}\mu a^2 \omega^2 \frac{\lambda}{2}$$

$$< E > = \frac{1}{8}\mu a^2 \omega^2 \lambda \qquad \ldots.(iv)$$

Also $\rho = \frac{\mu}{s}$ and $\lambda = \frac{2\pi}{K}$ pat in …(iv)

$$< E > = \frac{1}{4}\pi s \rho \frac{(a\omega)^2}{k} \qquad \text{Ans.}$$

Analysis :

on kinetic energy : from (i) :

$$dk = \frac{1}{2}\mu a^2\omega^2 \sin^2 kx \sin^2 \omega t\, dx$$

At node point : sinkx = 0
then dk = 0 (always at rech time)
At antinode point : sinkx = ±1

$$\text{then } dk = \frac{1}{2}\mu a^2\omega^2 \sin^2 \omega t\, dx$$

this equation suggest that kinetic energy will be
variable and it maximum when $\sin\omega t = 1 \Rightarrow \cos\omega t = 0$
then string particle will beat mean position.
On potential Energy from (ii)

$$dU = \frac{1}{2}\mu a^2\omega^2 \cos^2 kx \cos^2 \omega t\, dx$$

At node point : $\sin kx = 0 \Rightarrow \cos kx = \pm 1$
And hence potential energy will be maximum at node and minimum (0) at antinode always
At node :

$$dU = \frac{1}{2}\mu a^2\omega^2 \cos^2 \omega t\, dx$$

Potential energy at node will be maximum when
$\cos\omega t = 1$ and this instant all particles will be at extreme position

Analysis :

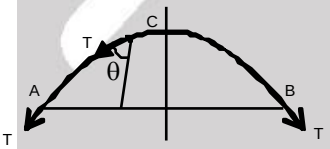
Energy between two node from (iii)

$$E = \frac{1}{2}\mu a^2\omega^2\left[\sin^2 \omega t \int_0^{\lambda/2} \sin^2 kx\, dx + \cos^2 \omega t \int_0^{\lambda/2} \cos^2 kx\, dx\right]$$

$$E = \frac{1}{2}\mu a^2\omega^2\left[\frac{\lambda}{4}\sin^2 \omega t + \cos^2 \omega t \frac{\lambda}{4}\right]$$

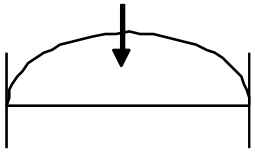
$$E = \frac{a^2\omega^2\lambda}{8}\mu = \text{const dues not depend on time.}$$

Energy is confined between two node.
Explanation with mechanics



Here point A and B Due node points which displace ment are zero. Hence work done by tension will be zero. And then energy will be conserved.
For point C : Tension component work and hence energy wil charge of particle and each half cycle work done by tension will Zero. And energy will be conserved in each half cycle.

Method : 2



Since enegy between two nodes are conserved at reach instant of time. Now calculate energy when string is at mean position and at mean position potential energy is zero. Hence total energy will be equal to kinetic energy at mean position.

At mean position :

E = U + K

Here U = 0

Now $K = \sum \frac{1}{2}(dm)v^2$

$K = \sum \frac{1}{2}(\mu dm)v^2$ …(i)

Now $v = \frac{dE}{dt} = -a\omega \sin kx \sin \omega t$

from (i) :

$K = \sum \frac{1}{2}(\mu dx)a^2\omega^2 \sin^2 kx \sin^2 \omega t$ …(ii)

At mean position $\varepsilon = 0 \Rightarrow \cos\omega t = 0 \Rightarrow \omega t = \frac{\pi}{2}$

from (ii)

$K = \sum \frac{1}{2}\mu a^2\omega^2 \sin^2 kx\, dx$

$\frac{1}{2}\mu a^2\omega^2 \int_0^{\lambda/2} \sin^2 kx\ dx = \frac{a^2\omega^2\mu\lambda}{8}$

$E = K = \frac{a^2\omega^2\mu\lambda}{8}$ Put $\mu = \rho s$ and $\lambda = \frac{2\pi}{K}$

$E = \frac{1}{4}\pi s\rho \frac{(a\omega)^2}{k}$